

Advanced Programming Practices

MTech 2026

Lab - 4

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP4_Qn.c (e.g. ABC2026001_APP4_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a C program to create a grade sheet using structures containing names, roll numbers, marks, and grades of 5 students. The program should be able to insert a new record, delete it, and modify it based on user input, i.e., menu-driven.
2. Write a C program to create an array having 10 elements and initialize it with numbers 1 to 10. Print the array. Take a pointer, say p, point to the base address, and loop through the addresses to access each element address and increase the value at the address of each element by 2. Again, print the elements of the array.
3. Take a 3-dimensional array, say a[p][r][c], where p denotes planes, each with r rows and c columns. Initialize the array with numbers sequentially from 1 to p*r*c, and take a pointer, x, pointing to the base address of the array. Write a C program to print the maximum element among the corner elements of each plane accessed through the pointer. (refer Q2 in Lab 2 assignment).
4. Using program 1, write a program to find the student having the highest mark.